1. **Question Format and Score: Crossover Design**. Three weeks ago you took your midterm exam. Unbeknownst to you, the exam was also serving as a crossover experiment. There were four versions (A, B, C, D) where versions A/C shared one structure and versions B/D shared another. The treatment of interest was question *format*: whether a question was presented as a single compound sentence or as separate sentences/bullet points. Questions 21 and 23 each had a compound and a separated version; the two structures (A/C vs B/D) ensured that if Q21 was compound then Q23 was separated, and vice versa.

   Exam versions were randomly assigned to students in blocks by session (morning or afternoon).

   a. What are the experimental units in this study? What are the measurement units?
   b. What is the treatment? How many levels does it have?
   c. Why is this a crossover design rather than a simple completely randomized design? What is the advantage of each student receiving both formats?
   d. What is the blocking factor, and why was blocking used?

2. **Coffee and Reaction Time: Crossover Design**. A researcher wants to know whether drinking coffee before a cognitive test improves reaction time. She recruits 20 subjects and uses a crossover design: each subject takes the cognitive test twice, once after drinking coffee and once after drinking decaf (a placebo). The order (coffee-first vs decaf-first) is randomly assigned, with a one-week washout period between sessions.

   a. What is the primary advantage of this crossover design compared to a completely randomized design that assigns 10 subjects to coffee and 10 to decaf?
   b. The researcher is concerned about a *carryover effect*: caffeine from the first session might still affect performance in the second session even after a week. Explain what a carryover effect is and why it is a threat to this crossover design.
   c. If carryover effects are present and asymmetric (i.e., the carryover from coffee to decaf is different from the carryover from decaf to coffee), explain why the crossover estimator $\bar{Z}$ (the average of the observed differences) from the previous problem would be biased. What assumption from the previous problem is violated?
   d. One way to test for carryover is to compare the *total* response (session 1 + session 2) between the two sequence groups. Explain the logic behind this test: under no carryover, why should the sequence groups have the same expected total?

3. **Coffee and Reaction Time: Analysis**. The researcher from the previous problem ran her crossover experiment and collected data. You can generate a synthetic version of her dataset with the following code.

```
library(tidyverse)
set.seed(40)
n <- 20
coffee <- tibble(
    subject  = 1:n,
    sequence = rep(c("coffee_first", "decaf_first"), each = n / 2),
    subject_ability = rnorm(n, mean = 250, sd = 30)
```

```r
) |>
    mutate(
        period1 = case_when(
            sequence == "coffee_first" ~ subject_ability - 8 + rnorm(n, 0, 10),
            sequence == "decaf_first"  ~ subject_ability + rnorm(n, 0, 10)
        ),
        period2 = case_when(
            sequence == "coffee_first" ~ subject_ability + 5 + rnorm(n, 0, 10),
            sequence == "decaf_first"  ~ subject_ability - 8 + 5 + rnorm(n, 0, 10)
        )
    ) |>
    select(subject, sequence, period1, period2)
```

Here, `period1` and `period2` are reaction times (in milliseconds) on the cognitive test in session 1 and session 2 respectively. We'll denote these Y_{i1} and Y_{i2} for subject i. Lower is better. The data-generating process includes a true treatment effect of coffee (coffee lowers reaction time by 8 ms) and a period effect (reaction time increases by 5 ms in period 2, perhaps due to fatigue or reduced novelty). There is no carryover effect.

a. For each subject, compute Z_i, the within-subject difference in reaction time (decaf − coffee). Be careful: which period corresponds to the coffee score depends on the subject's sequence. Add this column to the data frame and compute $\bar{Z}$.

b. Conduct a randomization test of the sharp null hypothesis that coffee has no effect on any subject's reaction time.

   i. State the null hypothesis in terms of potential outcomes.
   ii. Simulate 1,000 test statistics under the null by re-randomizing the sequence assignment (a complete randomization of 10 to each sequence).
   iii. Plot the null distribution with the observed statistic and report the two-sided p-value.

c. Reshape the data into long format (one row per subject per period) and fit a linear model with fixed effects for subject:

$$Y_{ij} = \beta_0 + \beta_1 \text{treatment}_{ij} + \beta_2 \text{period}_{ij} + \alpha_i + \varepsilon_{ij}$$

   where α_i is a fixed effect for subject i. Report and interpret $\hat{\beta}_1$. Compare this with the result from the randomization test.

d. Now fit a simpler model that omits the period effect: $Y_{ij} = \beta_0 + \beta_1 \text{treatment}_{ij} + \alpha_i + \varepsilon_{ij}$. How does the estimate of the treatment effect change? Explain why the crossover design protects the treatment effect estimate from period effects even when period is not included in the model.

Check back later for more questions.